

ADAI course portfolio

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Individual repository:

<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025>

Group repository:

<https://github.com/FontysVenlo/groupepository-group-2>

Learning goal I

Start Level: Beginning

I consider myself a beginner in machine learning and neural networks, with limited theoretical knowledge and practical experience. My initial experience involved developing a basic image labeling AI model, which introduced me to concepts of artificial neural networks (ANNs) and convolutional neural networks (CNNs) in a practical context. (<https://github.com/Jazziki17/A-systems/tree/main/src>)

Weekly progress and reflections

- **Week 1:** research some information about CNNs as well as other topics discussed with the group.
(<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/Resources/Research%20questions%20and%20answers.txt>)
Reflection: Due to a misunderstanding, I missed the lesson so I need to make sure to always check the schedule so I can attend consistently. I will make an effort to keep in contact with the group so I don't miss any information.
- **Week 2:** build a simple ai using DDI (diverse dermatology images) using TensorFlow.
(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions/aimodel.ipynb)
Reflection: I did not have very much material prepared for the meeting and was unable to contribute much to the discussion, I was asking questions and still was part of the conversation. For the upcoming weeks I need to adequately prepare the needed tasks and make sure I am prepared.
- **Week 3:** try again to build a CNN this time starting from a pretrained model.
(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/model.ipynb)
Reflection: As the scribe this week I was able to take some notes, however I considered my notes to not be descriptive/detailed enough so I need to focus more on the conclusion of the discussion. I will try to learn from the notes from my colleagues.
- **Week 4:** create an EDA(Exploratory Data Analysis) and do QA (quality assurance) for the presentation (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/EDA.ipynb)
Reflection: I was able to research and understand what is an EDA so I can successfully make one for the portfolio. I had to leave my QA task on the presentation till later in the

week so give me group time to add to it before I make changes, however this resulted in a lot of wasted time so I need to organize the tasks better with the group so this does not happen again.

- **Week 5:** help prepare the presentation

Reflection: I managed to discuss better task organization with the group and focus better on helping with the presentation. This week I lacked any progress on the building of the models and documenting the dataset so I need to remember to try and make more progress in this technical area. I will make task lists each week to better keep track of what I need to do

End level: Proficient

I consider to have achieved a good understanding of real-world application of machine learning, ANNs and CNNs, their requirements and limitations. I also consider to have deepened my understanding of the inner workings of neural networks and machine learning and I understand there is more to learn on this topic.

Learning goal II

Start level: Orienting

I have seen the importance of accurate labels when training machine learning models. I understand that good data labelling leads to better quality and consistency.

Weekly progress and reflections:

- **Week 10:** I set up label studio and did my set of labelling for the group assignment (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/Resources/data_labels.csv)

I also completed the individual exercise.

(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/annotation_exercise/annotation_exercise.ipynb)

Reflection: I think I managed all my work well and completed the assignments within the needed deadlines. I think I had a lack of communication with my group as we only spoke during the class hours. I think I need to keep working as I have, but I should take an initiative to start discussions in the group so we can work well together.

- **Week 11:** Helped put all our data together into one csv file: (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/annotation_exercise/data_combined.csv)

Did a simple analysis on our combined annotation data:

(https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/annotation_exercise/annotation_analysis.ipynb)

Reflection: While I thought I followed a good pattern for labelling, when I discussed with the team (and looked through the csv/analysis) I realised the importance of setting clear rules on these annotations. No wonder AI is not very good at labelling these images if we can't agree on the same labels.

End level: Beginning

I have applied a manual annotation technique using specialized software. I understand the importance of using multiple annotators to avoid bias. I have learned how to use and the importance of Cohen's Kappa when evaluating annotations in order to spot inconsistencies.

Learning goal III

Start level: Undefined

I had limited knowledge about Large Language Models (LLMs) and their practical applications in data science pipelines.

Weekly Progress and Reflections

- **Week 12:** LLM Research and Integration I explored how LLMs could be integrated into predictive pipelines, researching their appropriate use cases versus traditional predictive models.

(<https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/Resources/LLM%20Q&A.txt>)

Reflection: I learned that LLMs are better suited for explaining predictive model results rather than making predictions themselves

- **Week 13:** Initially did some testing on Gemini LLM to familiarize myself with usage. Afterwards we decided as a team that a predictive model would do the predictions and a LLM would interpret.

Testing LLM: https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/LLM_project/LLM_testroom/genai.py,
https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/LLM_project/LLM_testroom/context_test.py,
https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/LLM_project/LLM_testroom/prompting_numbers.py.

Sarima model predictions:

https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/LLM_project/SARIMA_model/model.ipynb

Reflection: From this I learned that, at a base level, an LLM is only ever good at sounding like a real person and rephrasing predefined ideas. Because of that it's best use case is as a middleman between a real person and a machine executing the needed task.

End level: Beginning

I have gained practical experience implementing LLMs and understand their role in making predictive model outputs more interpretable. I recognize the distinction between LLMs and traditional ML models, and understand basic considerations around proper context configuration.

Learning goal IV

Start level: Orienting

I had basic awareness of transfer learning and understood that pre-trained models could improve performance with limited data.

Weekly Progress and Reflections

- **Week 3:** Transfer Learning Implementation After struggling with training a CNN from scratch in Week 2, I implemented transfer learning using a pre-trained model for skin lesion classification. (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/model.ipynb)
Reflection: Using ResNet50 helped reduce training time compared to building from scratch.
- **Week 4:** Model Analysis Created an EDA to understand dataset characteristics and validate the transfer learning approach. (https://github.com/Razvan-Marian-Budurovici/ADAI-individual-repository-2025/blob/main/AIModel/skin_lesions_V2/EDA.ipynb)
Reflection: The EDA helped me understand the dataset better, though I still need to learn more about how these insights should guide model selection.

End level: Proficient

I have applied transfer learning in projects using pre-trained models, understanding when to leverage existing models versus training from scratch. I can implement basic transfer learning workflows and recognize the benefits when working with limited data.

Learning goal V

Start level: Beginning

I have participated in many projects before and have made use of skills like teamwork and investigative ability. I understand there are still areas that I lack and I strive towards improvement.

Week 1:

I missed the lesson due to a misunderstanding with the schedule but I contacted my group and tried to get up to speed with all the information I found on the course.

Week 2:

We worked in a group to discuss the previous week's research questions, I did not have everything researched myself so I tried to add to the conversation in other ways, like proposing questions or discussing vague terms.

Week 3:

I was assigned as the scribe this week so I mostly focused on writing down important notes from the meeting, however this week's discussion was pretty unorganized so it was hard to focus on both writing important notes as well as following the conversation.

(Document where I wrote the notes: https://github.com/FontysVenlo/groupepository-group-2/blob/main/ADAI_Skin_Cancer_Notes.md)

Week 4:

I was assigned QA for the presentation slides, however the day before the lesson the slides were still empty so I contacted the group to remind them to upload their information.

Week 5:

I discussed organization with the group to maybe help prevent leaving everything to the last day and organize the tasks with deadlines and minimum requirements. I also informed colleagues about changes to the presentation and some quality standards to follow.

Week 10:

I tried to organize the tasks with the group but as everyone was focused on the individual assignment, I decided to do the same and we would organize at a later time via online communication.

Weeks 11-13:

During these shorter following assignments with changing groups it was very hard to keep organized and efficient. During lecture hours I tried to keep everyone focused on the objective even if I was not the appointed chair. I adapted to the situation in order to try and avoid group work and focus on individual assignments in the group, as that seemed to work better at this point.

End level: Proficient

I have developed my ability to work in group settings by taking on roles such as scribe and quality assurance. I learned to address organizational challenges by initiating discussions about task management and deadlines. Through experience with changing group compositions, I recognize when to take initiative and when to adapt my approach based on team dynamics.